Requirements Matrix Draft

Use Case/ Functional Requirement	UI/Frontend	Dicom Parser Module	Segmentation Module	Rendering Engine	Marching Cubes Module	G-Code Generator Module
UC-1: Import Dicom File						
Allow user to select and upload a DICOM file or folder	File upload button, selection and drag-and-drop interface					
Validate DICOM format before processing	Display progress and errors	Validates DICOM files				
Parse DICOM data to extract voxel intensity and metadata		Reads slices, extracts voxel and metadata				
Notify user of success or failure	Pop-up status messages	Returns validation result				
UC-2: Select Anatomical Region						
Allow user to select tissue type (bone, skin, muscle)	Dropdown for tissue selection	Provides metadata for thresholds	Applies segmentation thresholds	Updates 3D preview dynamically		
Filter CT data based on selected tissue		Supplies voxel intensity data	Threshold filtering logic	Visualizes filtered data		
Update visualization automatically when region changes	Handles selection from user		Applies new mask	Rerenders visualization		
UC-3: Visualize 3D Anatomy						
Generate and display 3D preview	“Preview” button and controls	Provides volume data	Provides segmentation data	Renders model		
Allow user to rotate, zoom, and inspect	Camera and interactions controls			Real-time rendering		

model interactively						
Update visualization dynamically based on user input	Threshold sliders	Provides data	Recomputes segmentation	Updates rendered output		
UC-4: Generate STL File						
Allow user to select STL (.stl) output type	Dropdown file type selection					
Apply Marching Cubes to extract polygonal surface		Supplies Voxel Grid	Provides segmented mask		Surface Extraction	
Generate and export 3D mesh (.STL)	“Export STL” Button		Supplies final segmentation		Encodes STL Mesh	
Notify user when file generation is complete or failed	Pop-up status message				Reports Success / Error	
UC-5: Generate G-Code File						
Allow user to select G-Code (.gcode) output type	Dropdown file type selection					Initializes Module
Generate G-Code reproducing tissue density/opacity		Supplies Voxel/ Mesh data	Provides tissue density mapping			Converts to printer-ready G-code
Validate and save G-Code to user location	“Save G-Code” button					Performs Syntax Validation
Notify user when file generation is complete or failed	Pop-up status message					Performs validation